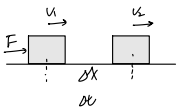
Energy



how much speed is accumulated when or force acts on a body

$x(t) = x(0) + v(0)t + \frac{1}{2}at^2$

$v(t) = v(0) + at \Rightarrow t = \frac{v(t) - v(0)}{a}$

$v(t)^2 - v(0)^2 = (at)^2 + 2at v(0) = 2a\left(x(t) - x(0)\right)$

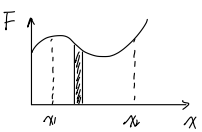
$v_2^2 - v_1^2 = 2a(x_2 - x_1) = 2\frac{F}{m}\Delta x$

$\frac{1}{2}mv_2^2 - \frac{1}{2}mv_1^2 = F\Delta x$

$\Delta k = k_2 - k_1 = F\Delta x$ newton x meter = Joule (J)

kinetic energy work done by F

$\frac{dk}{dt} = F\frac{dx}{dt} = Fv$ power : Joule/second = Watt



$\Delta k = \int_{x_1}^{x_2} F(x)\,dx$

$k_2 - k_1 = \int_{x_1}^{x_2} F(x)\,dx = G(x_2) - G(x_1)$ G. Se $\frac{dG}{dx} = F$

$k_2 - G_2 = k_1 - G_1$

$k_2 + U_2 = k_1 + U_1$ U Se $\frac{dU}{dx} = -F$ $dU = -F\,dx$

E_2 E_1 Conservation of Energy

when an objecs moves, its velocity changes (k), its position changes (U)
but some combination stay unchanged (k+U)

Eg. y ↑ 

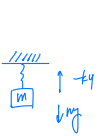
$F = -mg$ $\frac{1}{2}mv_2^2 - \frac{1}{2}mv_1^2 = \int_{y_1}^{y_2} -mg\,dy = mgy_1 - mgy_2$

$\frac{1}{2}mv_1^2 + mgy_1 = \frac{1}{2}mv_2^2 + mgy_2$

Eg. 

$\frac{1}{2}mv_2^2 - \frac{1}{2}mv_1^2 = \int_{x_1}^{x_2} -kx\,dx = -\frac{k}{2}x_2^2 + \frac{k}{2}x_1^2$

$\frac{1}{2}mv_1^2 + \frac{k}{2}x_1^2 = \frac{1}{2}mv_2^2 + \frac{k}{2}x_2^2$

Eg. 

$\frac{1}{2}mv_2^2 - \frac{1}{2}mv_1^2 = \int_{y_1}^{y_2} -mg - ky\,dy = -mg(y_2 - y_1) - \frac{k}{2}(y_2^2 - y_1^2)$

$E = \frac{1}{2}mv^2 + mgy + \frac{k}{2}y^2$

· Energy in High-Dim

$k = \frac{1}{2} m \|v\|^2$

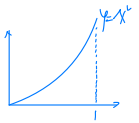
$\frac{dk}{dt} = m \langle v(t), a(t) \rangle = \langle v(t), F(t) \rangle$

$dk = \langle F(x,y), \begin{bmatrix} dx \\ dy \end{bmatrix} \rangle$

in high-dim, the work done also depends on the path

in addition to the starting and ending point

Eg. $F(x,y) = \begin{bmatrix} x^2 y^3 \\ xy^2 \end{bmatrix}$



$k_2 - k_1 = \int_\rho \vec{F}\, d\vec{s}$

$= \int_\rho \begin{bmatrix} x^2 y^3 \\ xy^2 \end{bmatrix} \begin{bmatrix} dx \\ dy \end{bmatrix}$

$= \int_0^1 \underbrace{x^2 x^6}_{y^3} + \underbrace{x x^4}_{y^2} \underbrace{2x}_{dy/dx}\, dx$

$= \int_0^1 (x^8 + 2x^5)\, dx$

Conservative Force : a force s.t. the integral doesn't depend on the path

construct conservative force

pick any potential $U(x,y)$

construct $F(x,y) = -\nabla U(x,y)$

$k_2 - k_1 = \int_1^2 F\, ds = \int_1^2 -\frac{\partial u}{\partial x} dx - \frac{\partial u}{\partial y} dy = \int_1^2 -du = u_1 - u_2$

$k_1 + u_1 = k_2 + u_2$

eg. the force $\begin{bmatrix} x^2 y^3 \\ xy^2 \end{bmatrix}$ is not a conservative force

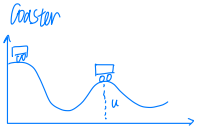
since it's not gradient of any function

Eg. Gravity



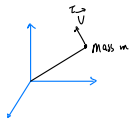
$F(x,y) = \begin{bmatrix} 0 \\ -mg \end{bmatrix}$ $U(x,y) = \begin{bmatrix} 0 \\ mgy \end{bmatrix}$

Eg. Roller Coaster



$mgy_{(0)} + \frac{1}{2} m v_{(0)}^2 = mgy_{(1)}$

Linear / Angular Momentum



Inertial linear momentum $\vec{P} = m\vec{V}$

$\frac{d}{dt}\vec{P} = m\vec{a} = \vec{F}$

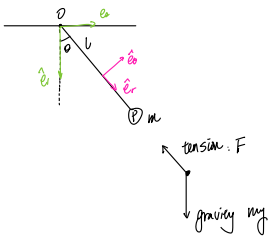
Special case: when $F=0$, linear momentum doesn't change

angular momentum of P wrt O: $\vec{h}_{P/O} = \vec{r}_{P/O} \times \vec{p}_{P/O} = m\,\vec{r}_{P/O} \times \vec{v}_{P/O}$

$$\frac{d}{dt}\vec{h}_{P/O} = m\frac{d}{dt}\left(\vec{r}_{P/O} \times \vec{v}_{P/O}\right)$$
$$= m\left[\vec{v}_{P/O} \times \vec{v}_{P/O} + \vec{r}_{P/O} \times \vec{a}_{P/O}\right]$$
$$= \vec{r}_{P/O} \times \vec{F}_P$$

$\vec{M}_{P/O} = \frac{d}{dt}\vec{h}_{P/O} = \vec{r}_{P/O} \times \vec{F}_P$ moment / torque on P about O

Eg pendulum



Inertial frame $\{O, \hat{e}_1, \hat{e}_2, \hat{e}_3$ pointing out the screen$\}$

polar frame $\{O, \hat{e}_r, \hat{e}_\theta, \hat{e}_3$ pointing out the screen$\}$

Inertial velocity: $\vec{v}_{P/O} = \frac{d}{dt}(r_{P/O})_I = \left(\frac{d}{dt}(r_{P/O})_B\right)_I + \omega \times (r_{P/O})_I = \dot{\theta}\hat{e}_3 \times l\hat{e}_r = \dot{\theta} l \hat{e}_\theta$

linear momentum: $\vec{P}_{P/O} = m\vec{v}_{P/O} = m\dot{\theta}l\hat{e}_\theta$

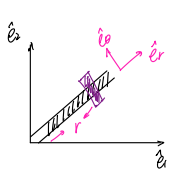
angular momentum: $\vec{h}_{P/O} = \vec{r}_{P/O} \times \vec{p}_{P/O} = l\hat{e}_r \times m\dot{\theta}l\hat{e}_\theta = ml^2\dot{\theta}\hat{e}_3$

$\frac{d}{dt}\vec{h}_{P/O} = \frac{d}{dt} ml^2\dot{\theta}\hat{e}_3 = ml^2\ddot{\theta}\hat{e}_3$

$\frac{d}{dt}\vec{h}_{P/O} = \vec{r}_{P/O} \times \vec{F}_P = (l\hat{e}_r) \times (-F\hat{e}_r + mg\hat{e}_1) = mgl(\hat{e}_r \times e_1) = -mgl\sin\theta\,\hat{e}_3$

$\ddot{\theta} = -g/l \sin\theta$

Eg. Ring on a rotating table



$\vec{v}_{P/O} = \dot{r}\hat{e}_r + r\dot{\theta}\hat{e}_\theta$

$\vec{a}_{P/O} = (\ddot{r} - r\dot{\theta}^2)\hat{e}_r + (2\dot{r}\dot{\theta} - r\ddot{\theta})\hat{e}_\theta$

$\ddot{r} - r\dot{\theta}^2 = 0$ no force in $\hat{e}_r$ direction

$\ddot{r} = r\dot{\theta}^2$